13 Achieving high availability: Availability zones, autoscaling, and CloudWatch

This chapter covers

- Recovering a failed virtual machine with a CloudWatch alarm
- Using autoscaling to guarantee your virtual machines keep running
- Understanding availability zones in an AWS region
- Analyzing disaster-recovery requirements

Imagine you run an online shop. During the night, the hardware running your virtual machine fails. Your users can no longer access your web shop until the next morning when you go into work. During the eight-hour downtime, your users search for an alternative and stop buying from you. That's a disaster for any business. Now imagine a highly available web shop. Just a few minutes after the hardware failed, the system recovers, restarts itself on new hardware, and your e-commerce website is back online again—without any human intervention. Your users can now continue to shop on your site. In this chapter, we'll teach you how to build a highly available system based on EC2 instances like this one.

Virtual machines are not highly available by default, so the potential for system failure is always present. The following scenarios could cause an outage of your virtual machine:

- A software problem causes the virtual machine's OS to fail.
- A software problem occurs on the host machine, causing the virtual machine to fail (either the OS of the host machine fails or the virtualization layer does).
- The compute, storage, or networking hardware of the physical host fails.
- Parts of the data center that the virtual machine depends on fail: network connectivity, the power, or the cooling system.